LANDTECH 2026 · P3 FRAUD & DOUBLE-SELLING DETECTION

# ShareTally
# অংশ-হিসাব

A plot cannot be sold twice if every share is accounted for — before mutation.



The red part: land that had already been sold.

Team [TEAM NAME] · [Student 1] · [Student 2]

Prototype on synthetic data

THE PROBLEM

# The khatian lags the deed — and a second sale slips through the gap

### Two offices

Deeds: Sub-Registry (Law Ministry). Khatian: AC (Land) mutation (Land Ministry). The rule book names this split as a root of disputes.

### Slow mutation

Only ~30% of new owners applied for mutation within 3 months in a 2018 trial across 46 upazilas.
BIT, 2020

### Weak safeguard

A sale deed is registered on the latest khatian plus the seller's own affidavit that they have not sold before.
Registration Act s.52A

> Until the first buyer mutates, the khatian still shows the seller — so the same share can be sold again, and the second buyer may reach the mutation desk first.

THE IDEA

# Treat every plot like a bank account of shares

1. The latest khatian **opens the account** — anna notation, decimals, acres become exact fractions.
2. Every registered deed **moves shares**, oldest first; earlier deeds keep priority (TPA s.48).
3. Moving more than is left **bounces** — with the earlier deeds as proof.
4. Missing statements make it **unconfirmed**, never "fine".

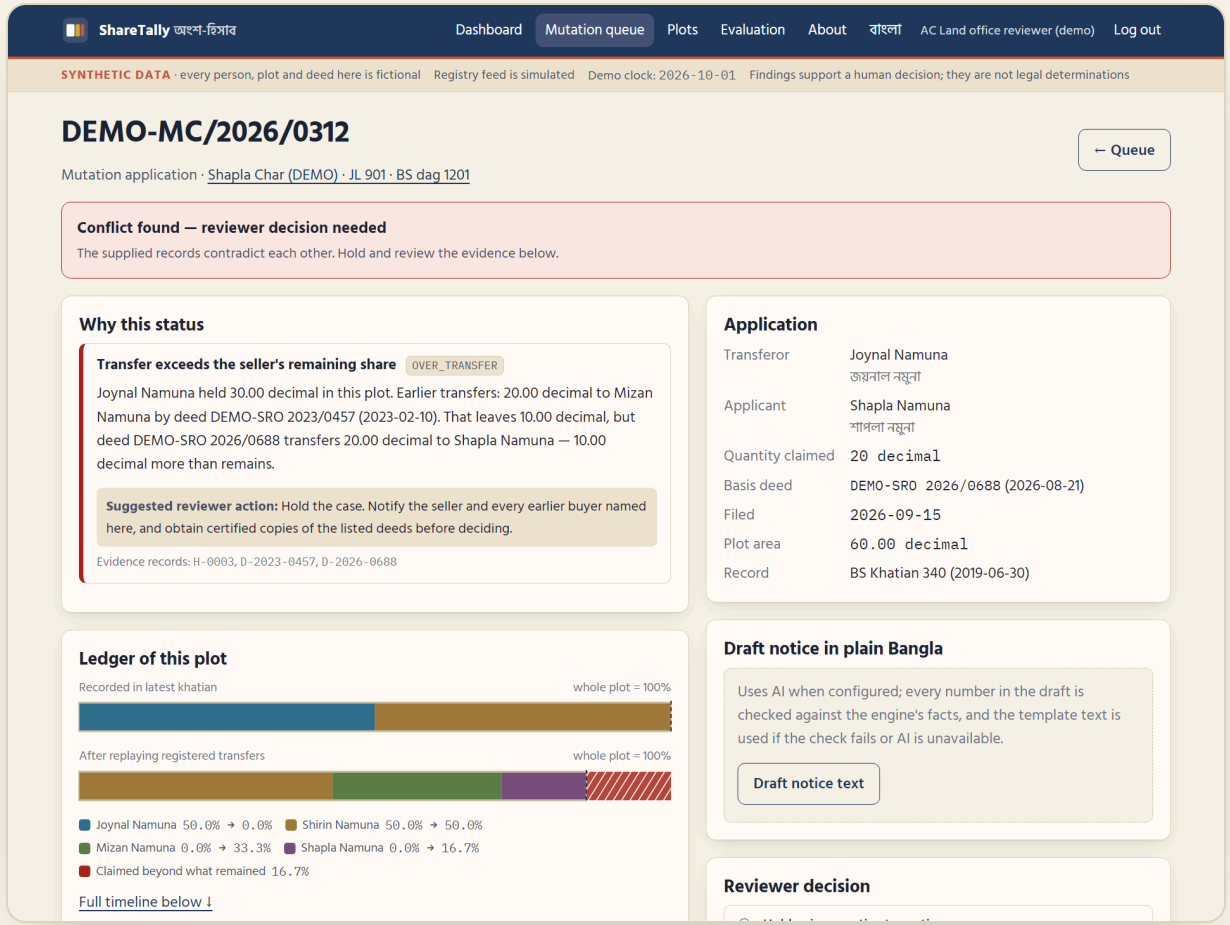
## The feed already exists in law

The Sub-Registrar must send a **Land Transfer (L.T.) notice** for every registered transfer (SAT Act 1950 s.89), and the land office must open a mutation file on it (s.143C).

We only need it **digital and reconciled**.

DEMO · WORKED EXAMPLE (FICTIONAL)

# Joynal held 30 decimal, sold 20 in 2023, now sells 20 again




**Conflict** `OVER_TRANSFER`

Remaining: **10.00** decimal

Claimed: **20.00** decimal

Shortfall: **10.00** decimal

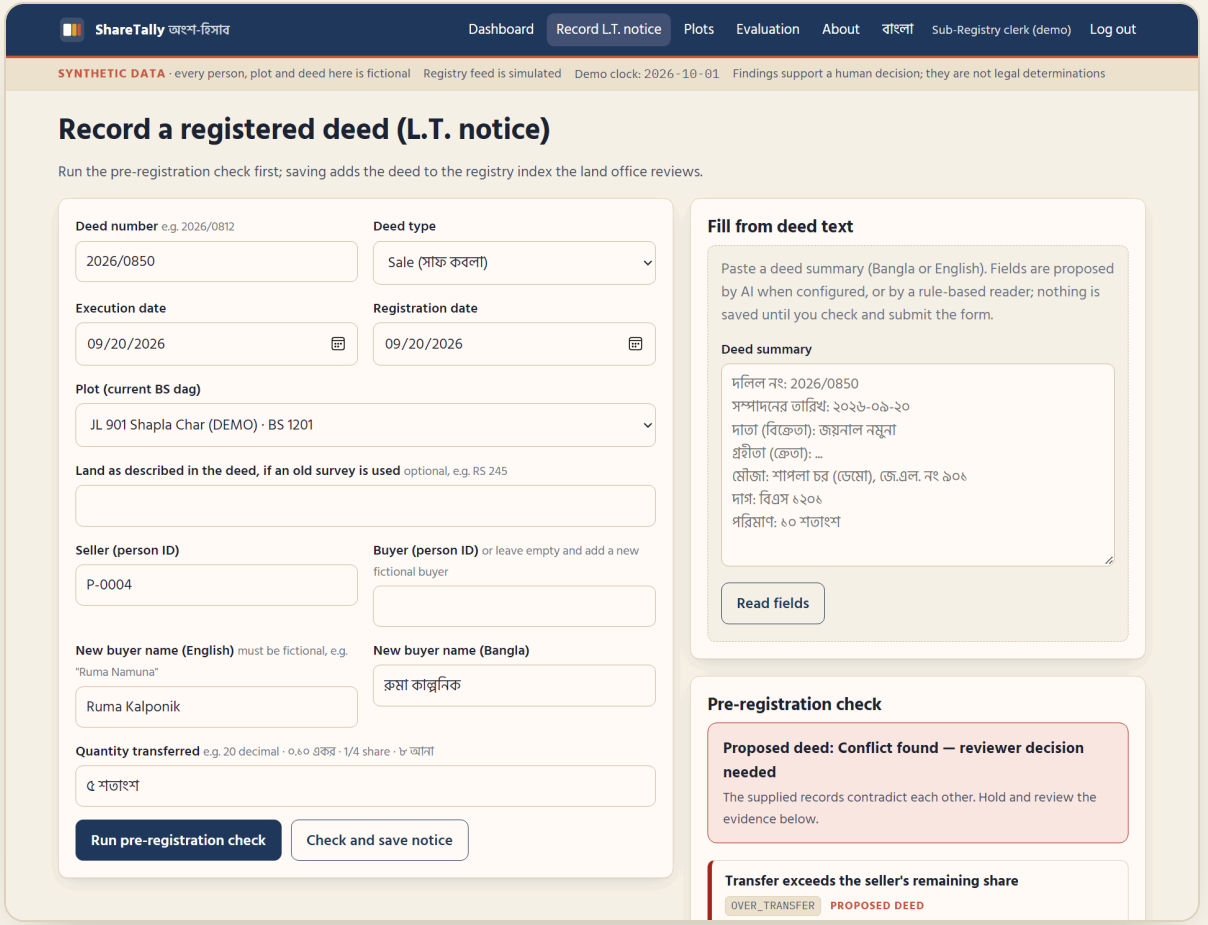
Earlier deed 2023/0457 was never mutated — so the khatian still showed Joynal. A khatian-only check would have cleared this case.

**Suggested action:** hold; notify every earlier buyer; get certified copies.

TWO USERS, ONE CHECK

# Catch it before registration, or before mutation

- **Sub-Registry clerk:** records the L.T. notice; **pre-registration check** before saving.
- **AC (Land) reviewer:** risk-sorted mutation queue, evidence, share strip, timeline, decision with reason.
- **Auditor:** read-only; verifies the tamper-evident log.
- Bangla / English; works on a phone.



HONEST BY DESIGN

# Missing data is never reported as clean

**Conflict**

Over-transfer · deed not in index · mutation ≠ deed · shares > 100%

**Incomplete**

Index gap · deed older than index · unknown seller · bigha undefined · old RS dag split · heir shares ≠ 1

**Attention**

Unmutated > 90 days · quick resale — context only

**No conflict**

"in the supplied records and coverage" — not a title guarantee

> ShareTally never decides ownership and never calls anyone fraudulent. The AC (Land) decides; "proceed" is blocked while a conflict or gap remains.

ARCHITECTURE & SAFEGUARDS

# Simple stack, auditable rules

## Stack

Python `fractions` ledger engine (pure, tested) · FastAPI · SQLite · server-rendered pages, vanilla JS · 51 automated tests · optional Claude API with offline fallback

## Security (own prototype only)

Hashed passwords · roles · CSRF · login throttling · strict CSP · hash-chained audit log storing the engine version with each decision

## AI, fenced in

Drafts a plain-Bangla notice; rejected if it adds any number not in the engine's facts or uses accusatory words. Names are tokenised before sending.

## Data

100% synthetic: 49 plots, 163 fictional people ("Namuna", "Kalponik"), 62 deeds. No portal scraped, no real record used.

EVALUATION (SYNTHETIC)

# Measured against results declared in advance

**70/70**
declared statuses and codes matched

**0/30**
clean plots wrongly flagged

**0/13**
incomplete cases called clean

**8/12**
problem cases a khatian-only check would clear

> Same team wrote rules and expectations: this proves the mechanism works as designed, not real-world accuracy. Next: blind cases from land officers, then an approved anonymised sample.

COUNTRYWIDE PATH

# Start from the notice the law already requires

### 0 · Validate

Blind tests and rule review with land officers and legal advisers

### 1 · Pilot

Digital L.T. notices from 1–2 Sub-Registries into one AC (Land) office; read-only check

### 2 · Integrate

API inside the e-mutation screen; pre-registration check at Sub-Registry

### 3 · Scale

Backfill historic deed indexes; upazila by upazila; per-plot compute scales out

**Main risk:** inter-ministry data sharing. **Cost:** prototype runs free; the check takes < 1 ms per plot on a laptop.

TEAM & ASK

# Every share accounted for.

Students: Tanvir Hossin [role] · [NAME]

Faculty mentor: [NAME]
Industry mentor: [NAME]

**Our ask:** one upazila pilot — digital L.T. notices from its Sub-Registry, ShareTally beside e-mutation, measured for three months.

Repository [link] · Live demo [link]